MACHINE LEARNING FOUNDATIONS · BEFORE YOU BEGIN

Set Up Your Machine Learning Workshop

About an hour of setup to do **before the first session**: Python 3.12 or newer, a virtual environment with the course packages, a notebook tool, and the Pinecrest Outfitters data. These are the same steps as Lesson 0.1 on the course site (rays-python-machine-learning.netlify.app, lesson "Your ML Toolkit"), so if you get stuck, that page shows every step with its expected output. Session 1 then starts with models instead of installers.

⏱️ **~60 min**, mostly downloads | Assumes Python + pandas | Windows · Mac · Linux | No GPU needed

> **PLEASE DO THIS AT HOME, NOT IN CLASS**
>
> The install downloads several hundred megabytes (the finished environment takes about 2 GB of disk). Venue Wi-Fi is rarely up to twenty people doing that at once. If a step fails, keep going with the others and bring the **exact error message** (a screenshot is fine) to Session 1: its first half hour includes an environment check with help.

WHAT YOU NEED

- **A laptop** (Windows, macOS or Linux) and its **charger**, with **permission to install software** and about **2 GB of free disk space**. Any ordinary processor will do: the course never needs a graphics card. Chromebooks, tablets and locked work laptops cannot run the course comfortably: talk to your instructor first.
- **Python 3.12 or newer** (64-bit), a **terminal** (PowerShell on Windows, Terminal on macOS), and **JupyterLab** (installed with the course packages) or **VS Code**.
- **Two downloads** from the course home page: `requirements.txt` and `pinecrest-ml-data.zip` (1.4 MB).
- `check_environment.py`, which your instructor sends with this guide.
- **A learning journal**: a plain text file called `journal.md` or a paper notebook. Every lesson ends with a journal prompt.

AM I READY FOR THIS COURSE?

Machine Learning Foundations follows **Python Data Science** (free at rays-home.netlify.app), or equivalent experience with Python and pandas. No statistics or machine-learning background is needed. You are ready if

you can, without looking things up:

- ☐ Slice a list (`scores[2:5]`, `scores[-2:]`) and build and loop over a dictionary
- ☐ Write a function with parameters, a default value and a `return`, and call it with a keyword argument
- ☐ Load a CSV with `pd.read_csv` and look at it with `head()`, `shape` and `dtypes`
- ☐ Select columns and filter rows with a condition (`df[df["spend"] > 100]`)
- ☐ Compute a rate per group with `groupby(...)[...].mean()`
- ☐ Find and count missing values with `isna().sum()`, and say what `mean()` does with them
- ☐ Read a traceback and find the line that caused it

Two or more unchecked? Review them in Python Data Science (Module 2, pandas) before Session 1. Session 1 includes a short, ungraded diagnostic on these same skills, so your instructor knows where to help.

SET UP IN 6 STEPS

1 Python 3.12 or newer

~5 min

Open a terminal and check what you have. A higher number than 3.12 is fine.

SYSTEM	CHECK	MISSING OR OLDER THAN 3.12?
Windows	`py --version`	Install the 64-bit release from **python.org/downloads**. In the installer, tick **"Add python.exe to PATH"** on the first screen. No administrator rights? The installer's "install for me only" option usually works without them.
macOS	`python3 --version`	The macOS installer from python.org, or with Homebrew: `brew install python@3.12`
Linux (Debian, Ubuntu)	`python3 --version`	`sudo apt install python3 python3-venv python3-pip`. Ubuntu 24.04 and newer give you 3.12. On an older release that still says 3.10 or 3.11, install `python3.12` and `python3.12-venv` (for example from the deadsnakes PPA) and use `python3.12` wherever Step 2 says `python3`.

2 A project folder and a virtual environment

~5 min

Everything for the course lives in one folder, `pinecrest-ml`, with its own environment inside it. Run these one line at a time:

	WINDOWS (POWERSHELL)	MACOS / LINUX
Folder	`mkdir pinecrest-ml` then `cd pinecrest-ml`	
Create	`py -m venv .venv`	`python3 -m venv .venv`
Activate	`.venv\Scripts\Activate.ps1`	`source .venv/bin/activate`
Check	Your prompt now starts with `(.venv)`. From here on, plain `python` works on every system.	

> **ACTIVATION IS PER TERMINAL WINDOW**
> Every time you sit down to work: open a terminal, `cd pinecrest-ml`, activate, look for `(.venv)`. Forgetting is the number-one cause of `ModuleNotFoundError`.

3 Install the course packages

~10–20 min

Download **requirements.txt** from the course home page into `pinecrest-ml`. It pins every package to the release the lessons were tested with (scikit-learn 1.9, pandas 3.0, XGBoost, LightGBM, imbalanced-learn, Optuna, JupyterLab, and the packages the Advanced kit uses later, such as PyTorch and MLflow), so your output matches the pages.

Linux only, first: install the CPU build of PyTorch. The default Linux download adds several gigabytes of graphics-card libraries the course never uses. On Windows and macOS, skip this line.

```
python -m pip install torch~=2.14.0 torchvision~=0.29.0 --index-url https://downlo
```

Everyone: with `(.venv)` showing, install the whole list:

```
python -m pip install -r requirements.txt
```

Allow ten minutes or more. Success ends with a line starting **Successfully installed**. If you already use the `uv` tool, `uv venv` and `uv pip install -r requirements.txt` give the same result.

4 The course data, and where the labs go

~3 min

Download **pinecrest-ml-data.zip** (1.4 MB) from the course home page into `pinecrest-ml`, then unzip it there. This works on every system; the `.` means "extract here":

```
python -m zipfile -e pinecrest-ml-data.zip .
```

You now have a `data/` folder with seven CSV files, a `README.md` that describes every column, and `make_ml_data.py`, the program that generated the data. Pinecrest Outfitters is fictional: every customer, product, review and payment is invented.

```
pinecrest-ml/
├── .venv/              the environment (never edit by hand)
├── data/               customers_churn.csv, weekly_demand.csv, transactions.csv,
├── labs/               handed out in Session 1: goes here, next to data/
└── requirements.txt
```

In Session 1 your instructor hands out a `labs/` folder. Put it inside `pinecrest-ml`, next to `data/`, exactly as above: every lab finds the data by itself from there. Keep the file `labs/labs_paths.py`; every lab uses it.

5 JupyterLab or VS Code

~5 min

- **JupyterLab** (already installed): from `pinecrest-ml` with the environment active, run `jupyter lab`. A browser tab opens; click **Python 3 (ipykernel)** to make a notebook. Stop it with **Ctrl+C** in the terminal.
- **Or VS Code**: install the **Python** and **Jupyter** extensions, **File ▸ Open Folder** on `pinecrest-ml`, and pick the kernel and interpreter that mention `.venv`.
- Either way, the notebook must run on the `.venv` Python. If it runs on another Python, `import sklearn` fails even though you installed it.

6 Run the environment check

~2 min

Save `check_environment.py` (sent with this guide) in `pinecrest-ml`, next to `data/`, and run it with the environment active:

```
python check_environment.py
```

It checks four things: Python is 3.12 or newer, an environment is active, every course package imports (with its version), and the data folder has all seven files with the right number of rows. **The first run can sit silent for up to a minute** while PyTorch and MLflow import for the first time; the second run is much faster. The end of a good run looks like this (lines shortened with …):

```
3. Course packages
  [ok]   numpy                 2.5.4
  [ok]   pandas                3.0.6
  [ok]   scikit-learn          1.9.1
  …
  [ok]   PyTorch device        CPU only (expected)
4. Course data
  [ok]   7 data files          all present
  [ok]   row counts            customers_churn.csv 6,000 customers, and the rest
============================================================
Note: the deep-learning lessons (Module 10) download Fashion-MNIST, EuroSAT and p
      the first time they need them (about 175 MB). Nothing to do now.
READY for Python Machine Learning
```

Your last version digits may be higher. The note about images is for the Advanced kit: **nothing in Machine Learning Foundations downloads anything**.

✅ YOU'RE READY

Last line **READY for Python Machine Learning**? Reply to your instructor with that line, and you are set for Session 1. **NOT READY**? It prints a numbered list of exactly what to fix: try those and the table below, run it again, then send the whole output.

COMMON INSTALL PROBLEMS

YOU SEE…	TRY THIS
Windows: `python` is not recognized, or it opens the Microsoft Store	Before an environment is active, Windows only knows `py`. Create the environment with `py -m venv .venv`; once it is active, `python` works.
PowerShell: "running scripts is disabled on this system"	Run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` once, answer Y, and activate again.
Linux: pip starts downloading files named `nvidia-…` or `…cuda…`, several GB in all	pip chose the graphics-card build of PyTorch. Press Ctrl+C, run the CPU line from Step 3 first, then the requirements line again.
Debian/Ubuntu: "ensurepip is not available"	`sudo apt install python3-venv python3-pip` (or `python3.12-venv`), delete the half-made `.venv` folder, create it again.
pip: "Could not find a version that satisfies the requirement torch~=2.14.0" (or another package)	Check `python --version`: the course needs 64-bit Python 3.12 or newer. Install it, delete `.venv`, then repeat Steps 2 and 3. A very new Python that the pinned versions do not cover yet: install 3.12 alongside it and use that.
macOS: importing `xgboost` or `lightgbm` fails with a message about `libomp`	Install the OpenMP library with Homebrew: `brew install libomp`, then run the check again.
`ModuleNotFoundError: No module named 'sklearn'`	The environment is not active in this terminal (no `(.venv)`). In a notebook: the kernel is another Python; start `jupyter lab` from the activated terminal, or choose the `.venv` kernel in VS Code.
`No module named 'scikit_learn'`	The install name is `scikit-learn`; the import name is `sklearn`.
`FileNotFoundError` for `data/customers_churn.csv`, or the check says the data folder was not found	Python is working in another folder, or the zip was unpacked into a nested folder. You need `pinecrest-ml/data/customers_churn.csv`; run from `pinecrest-ml`.
The check prints nothing for 30 seconds or more	Normal on the first run (large packages build caches). Wait; the second run is much faster.
`jupyter lab` starts but no browser opens	Copy the `http://localhost:8888/lab?token=…` line from the terminal into your browser.
The install is slow or stops halfway	Run the same command again on a better connection: pip skips what it already has.

Environment badly broken? Delete the `.venv` folder and repeat Steps 2 and 3. Your notebooks and data are not inside it, so nothing of yours is lost.

THE COURSE AT A GLANCE

SESSIONS	YOU'LL LEARN
1–2	Your toolkit, the scikit-learn estimator API, honest splits, baselines, leakage and cross-validation
3–5	Metrics and thresholds in dollars; imputing, scaling and encoding; pipelines; creating and selecting features; rare classes
6	Linear and logistic regression, regularization and bias-variance (Checkpoint 1)
7–8	k-nearest neighbors, Naive Bayes, support vector machines, decision trees
9–10	Random forests, gradient boosting, voting and stacking, fair model selection; you choose your final project
11	Hyperparameter tuning, then a final-project studio and showcase

11 sessions of 3 hours. Every lesson uses the same fictional company, Pinecrest Outfitters: its customers, weekly demand, payments and reviews. Between sessions you finish each lesson's project, its exercises, quiz and journal entry on the course site: about two to three hours a week.

FREQUENTLY ASKED

DO I NEED A GRAPHICS CARD (GPU)?

No. Every lesson runs on an ordinary laptop processor. If you do have one, nothing changes: the lessons still run on the processor, so your numbers match the pages.

CAN I USE ANACONDA, COLAB OR A SETUP I ALREADY HAVE?

Yes, if you end up with Python 3.12 or newer and the versions in `requirements.txt`, and the check says READY. Demos in class use a `.venv`. A hosted notebook also needs the data folder uploaded, and its package versions may differ, so some numbers will not match the pages.

WHY DOES THE INSTALL INCLUDE PYTORCH IF THIS KIT NEVER USES IT?

One environment serves the whole course, including the Advanced kit's deep-learning lessons, and the environment check expects all of it. Install it once now and you never have to change the environment later.

IS THE PINECREST DATA REAL?

No. It was generated for the course (`data/make_ml_data.py` rebuilds it) to behave like real retail data, rare fraud and leaky columns included.

WHAT IF A STEP FAILS AND I CANNOT FIX IT?

Come anyway, with the output of `check_environment.py`. Session 1 opens with an environment check and help, and you can pair with a neighbor for the day.

FOR THE FACILITATOR: ROOM SETUP

This section is for the instructor; it can stay on the learners' copy or be left off when you print it.

- ☐ **A week before Session 1:** send this guide with `labs/module-01/l0_1-orientation/check_environment.py` attached; ask each learner to reply with its last line. Keep a list: no reply, or NOT READY, means an install risk to chase before the day.
- ☐ **USB sticks or a shared folder** (in case the network fails): `requirements.txt`, `pinecrest-ml-data.zip`, and the `labs/` folder **without any** `*_solution.py` files (keep `labs_paths.py` in it). Hand out each session's `labs/module-NN/` folder on the day, also without solutions.
- ☐ **Your own machine:** the same setup in your own `pinecrest-ml`; `python labs/module-01/l0_1-orientation/check_environment.py` ends with READY (so you can show what it looks like). From the kit folder, `python3 labs/check_labs.py module-01` prints PASS on every line; run the session's check before each session.
- ☐ **Projector:** a terminal with `(.venv)` active, JupyterLab or VS Code started from `pinecrest-ml`, font size 18 pt or more, the session's deck from `slides/` in a browser tab, and the lesson pages open.
- ☐ **The room:** a power socket for every laptop (bring power strips), a whiteboard, sticky notes (help flags and exit tickets), printed workbooks or a shared PDF, and Wi-Fi for browsing the lesson pages. No session in this kit needs a download or an account.
- ☐ **Install trouble on the day:** do not debug one machine in front of the room. Two minutes per machine, then pair the learner with a ready neighbor and fix it at the break or at home. Activation and the data folder are one-minute fixes; a full package install is not done on venue Wi-Fi. The Session 1 facilitator guide has the full trap table.
- ☐ **Locked work laptops:** the python.org installer's "install for me only" option usually needs no administrator rights; otherwise the learner's IT team installs Python 3.12 before Session 2.

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